Do Memory-Based Observers Forecast Further? A Matched Pilot Study of Long-Horizon Activation Forecasting in a Frozen LLM

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Abstract

Runtime monitors for long generations must warn before harm appears, yet activation readers such as probes and activation verbalizers answer only about the present or the next few tokens. We ask whether an *observer* with trajectory memory, learned future-window queries, a future-activation auxiliary loss and a censoring-aware hazard head forecasts further ahead than a matched direct readout of the latest activation. We compare three such observers (horizon-query, latent rollout, hierarchical coarse-to-fine) against a parameter-, input- and epoch-matched direct readout on a frozen QWEN3-8B, with horizons of 64 to 2048 tokens and a prompt-family-level train/test split. In a pilot of 84 on-policy trajectories, naturally occurring unsafe onsets are too rare (5 trajectories) and too early ($\leq$448 tokens) to test long-horizon safety forecasting; a text-only TF-IDF baseline is best on this event. On a denser "caution" event (first refusal or warning), the hierarchical observer reaches AUROC 0.976–0.990 versus 0.871–0.953 for the direct readout ($+0.04$ to $+0.11$), but family-bootstrap confidence intervals touch zero, the gain peaks at 256–512 tokens rather than growing with lead distance, and much of it reflects prompt-type separation. Memory is the component with the clearest long-horizon benefit (removing it from the rollout observer drops AUROC at 1024 tokens from 0.944 to 0.831); the future-activation loss, hazard head and latent rollout show no benefit. For long-form content, all observers forecast future-segment behavior at 0.83–0.88 accuracy up to 1024 tokens ahead and retrieve the remaining-response summary at 4–5$\times$ chance MRR. Our results suggest that trajectory memory is worth its cost, and that testing long-horizon unsafe forecasting requires data with late natural onsets.

1 Introduction

A monitor that flags harmful text only after it appears cannot stop it. This matters more as language models write long reports, run multi-step agents and produce thousands of tokens per response. A useful monitor would read the model while it generates and say, early, *when* the response is likely to turn unsafe and *what* the rest of the response will do.

Activations are a natural input for such a monitor. Probes recover global properties of an upcoming response, such as its length, from the last prompt state [Dong et al., 2025]. Activation verbalizers such as LatentQA [Pan et al., 2024] and Activation Oracles [Karvonen et al., 2025] read a fixed window of activations and answer free-form questions about the model, including properties of its future completion. Streaming safety monitors [Li et al., 2025, Kavumba et al., 2026, Qwen Team, 2025] act on partial generations.

What is missing? Every one of these readers answers about the present or about the whole next reply, without a time axis. The longest activation-based lookahead we found in the literature is about 32 tokens [Xie et al., 2026], and next-token-lens methods decay within a few tokens [Pal et al.,

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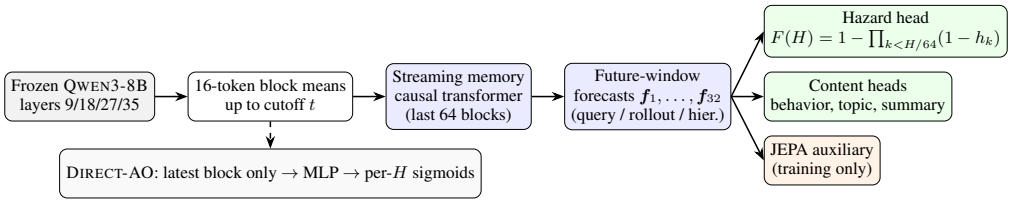


Figure 1: Observer pipeline. A frozen QWEN3-8B generates a response; activations from four layers are averaged over 16-token blocks. The observer reads only blocks up to the cutoff t, keeps a streaming memory, forms one forecast vector per future 64-token window, and outputs a censoring-aware hazard over 32 windows plus content predictions. A JEPA-style loss aligns forecasts with the target model's own future-window activations during training. The matched baseline DIRECT-AO reads only the latest block.

2023]. None keeps a memory of the growing activation trajectory, and none outputs a distribution over *when* an event occurs. Streaming safety monitors evaluate onsets that mostly occur within the first sentence and do not report performance by lead distance.

Our approach. We hypothesize that an observer with three properties forecasts further ahead than a direct readout trained on the same labels: (a) a streaming memory over the activation trajectory, (b) predicted latents for future windows, trained with a JEPA-style future-activation loss, and (c) an explicit discrete-time hazard head that handles right-censoring. We build three observers with these properties (figure 1): a horizon-query bottleneck (A-QUERY), a latent rollout (B-ROLLOUT) and a hierarchical coarse-to-fine model (D-HIER). We compare them against DIRECT-AO, a matched readout of the latest activation with the same encoder, inputs, targets, parameter budget (≈6M) and epochs, on a frozen QWEN3-8B. We evaluate two tasks: forecasting the first unsafe (or caution) segment within $H \in \{64, \ldots, 2048\}$ tokens, and forecasting the behavior, topic and summary of future segments of long responses.

What do we find? The answer is weak and mixed. Natural unsafe onsets on unmodified benchmark prompts are rare and early (5 of 84 trajectories, all $\leq$448 tokens), so the primary safety event is unpowered and a text-only baseline wins on it. On the denser caution event, D-HIER improves AUROC over DIRECT-AO by $+0.04$ to $+0.11$ at every $H \leq 1024$ (0.976–0.990 vs. 0.871–0.953), but confidence intervals touch zero, the gain peaks at $H = 256$–512 rather than growing with H, and a linear probe is nearly as good. Removing memory is the ablation that hurts most at long horizons; the future-activation loss, the hazard head and latent rollout bring no gain. On long-form content, observers forecast future-segment behavior at 0.83–0.88 accuracy up to 1024 tokens ahead (majority 0.62) and retrieve the remaining-response summary at 4–5$\times$ chance MRR.

Our contributions are:

- We formulate long-horizon activation forecasting as censored event-time prediction over lead distances up to 2048 tokens, with leak-checked inputs and a prompt-family split fixed before generation.
- We design three observer architectures that combine streaming memory, future-window queries, a future-activation auxiliary loss and a logistic hazard head, and compare them against a matched direct readout and five non-neural baselines.
- We run component ablations and zero/shuffled-activation controls, and show that memory is the one component with a consistent long-horizon benefit while the hazard head hurts long-horizon calibration under heavy censoring.
- We report the limits of natural data honestly: unmodified prompts give too few late unsafe onsets, and pooled AUROC largely measures prompt-type separation rather than timing.

Section 2 reviews related work, section 3 describes the task and observers, section 4 reports results, and section 5 discusses limitations.

2 Related Work

Activation verbalizers. Patchscopes [Ghandeharioun et al., 2024] and SelfIE [Chen et al., 2024] decode activations by patching them into a model's own forward pass. LatentQA [Pan et al., 2024] trains a LoRA decoder to answer questions about properties of future completions from current prompt activations, and outperforms linear probes and untrained patching on persona identification and relation extraction. Activation Oracles [Karvonen et al., 2025] scale this to about one million training examples across several models, including QWEN3-8B, and generalize to auditing tasks out of distribution. Predictive Concept Decoders [Huang et al., 2025] pretrain the decoder to predict the next 16 tokens, which improves detection of upcoming harmful output. All of these read a static window and answer without a time axis. Unlike them, we give the reader a memory of the whole trajectory and ask it to say *when* an event occurs, up to 2048 tokens ahead. Our DIRECT-AO baseline is a matched classification-head stand-in for these decoders, not an LLM decoder itself.

Lookahead in hidden states. Future Lens [Pal et al., 2023] finds that single hidden states predict tokens 1–3 positions ahead, with accuracy that decays quickly. Probes on the last prompt state predict global response properties such as length and answer choice [Dong et al., 2025], and progressive remaining-length prediction improves during decoding over up to 32 steps [Xie et al., 2026]. Multi-token prediction [Gloeckle et al., 2024] and the Belief State Transformer [Hu et al., 2024] study how much future information a model encodes. Building on these findings, we measure forecasting accuracy as a function of lead distance, at distances an order of magnitude longer.

Streaming safety monitors. Constitutional Classifiers [Sharma et al., 2025], Qwen3Guard-Stream [Qwen Team, 2025] and the Streaming Content Monitor [Li et al., 2025] classify partial generations token by token. StreamGuard [Kavumba et al., 2026] frames moderation as forecasting and supervises a value head with Monte Carlo rollouts. Survival framings of unsafe sampling [Davidov et al., 2025] count resampling attempts rather than tokens. Shallow safety alignment [Qi et al., 2024] shows that alignment acts mostly on the first few tokens, which explains why natural onsets are early. These monitors read text or token-level states and do not report performance by lead distance; we use activations, model censoring explicitly and stratify every metric by horizon.

Future-latent objectives and survival analysis. Contrastive predictive coding [van den Oord et al., 2018], next-latent prediction [Teoh et al., 2025], hierarchical latent prediction [Shi et al., 2026] and JEPA-style objectives [Assran et al., 2023] train representations to predict future embeddings. Discrete-time survival networks [Gensheimer and Narasimhan, 2018] model event times with a logistic hazard per time bin and a likelihood that handles right-censoring; Cox-Time and DeepHit [Kvamme et al., 2019, Lee et al., 2018] are alternatives. We reuse both ideas as observer components: a cosine loss to future-window activations and a censored logistic hazard over 64-token bins.

3 Method

3.1 Task

Target model and trajectories. We generate one response per prompt with a frozen QWEN3-8B in non-thinking mode (bf16, temperature 0.7, top-p 0.9, at most 2048 new tokens, seed 42). The observer never touches generation. After generation we recompute activations with one causal forward pass over prompt and continuation, and record the outputs of decoder layers 9, 18, 27 and 35.

Prompts and split. We use 96 unmodified benchmark prompts: 32 long-form prompts from LONGWRITER [Bai et al., 2024], 16 toxic user prompts from TOXICCHAT [Lin et al., 2023], 16 unsafe-category prompts from BEAVERTAILS [Ji et al., 2023], 16 unsafe prompts from AEGIS-2.0 [Ghosh et al., 2025], and 16 safe and contrast prompts from XSTEST [Röttger et al., 2024]. A prompt family is a source $\times$ category pair (16 hash buckets for LONGWRITER). We assign 30% of families to test *before* generation and assert that no family appears in both splits. Twelve responses shorter than 64 tokens give no cutoff, which leaves 84 trajectories (58 train, 26 test).

Annotation and events. An LLM annotator (gpt-5.6-luna) receives each response cut into 64-token segments and returns, per segment, a behavior label in {explain, instruct, recommend, narrate, refuse, warn, other}, a short topic, an unsafe flag and a short description, plus an overall summary.

We define two events: *unsafe*, the first segment flagged unsafe, and *caution*, the first segment labeled refuse or warn. Onsets are therefore known at 64-token resolution.

Prefix-level targets. Cutoffs are at $t = 64c$ for $c = 0, \ldots, 31$. For a cutoff c, hazard bin k is future segment $c + k$, giving 32 bins that reach 2048 tokens. Responses that end without the event are right-censored at their length rather than counted as safe. Safety training and evaluation use only still-safe prefixes (onset segment $\geq c$). Content targets are the behavior and topic cluster of each of the next 16 segments, where topics are mapped to 16 k-means clusters of MiniLM [Reimers and Gurevych, 2019] embeddings fit on train topics only, and the MiniLM embedding of the concatenated descriptions of all remaining segments (the remaining-response summary).

Observer inputs and causality. Each 16-token block is represented by the mean activation of the four layers, standardized with train statistics (16,384 dimensions). Block 0 is the mean of the last 8 prompt tokens and block j covers generated tokens $[16(j-1), 16j)$. At cutoff t the observer reads blocks $0, \ldots, t/16$; the input builder asserts that no later block is read. Because the forward pass is causal, these activations depend only on tokens up to t. Future blocks enter only as targets of the auxiliary loss.

3.2 Observers

All models share an input encoder (LayerNorm and a linear map $16384 \to 256$, about 4.2M of the $\approx$6M parameters), the same targets, 8 epochs of AdamW (learning rate 3×10^{-4}, batch 64) and 3 seeds.

Streaming memory. A-QUERY, B-ROLLOUT and D-HIER encode the last 64 blocks (1024 tokens) with a causal two-layer transformer, updated every 16 tokens.

Forecast mechanisms. A-QUERY uses 32 learned queries, one per future 64-token window, which cross-attend to the memory to produce forecast vectors $\boldsymbol{f}_1, \ldots, \boldsymbol{f}_{32}$. B-ROLLOUT rolls a latent $\boldsymbol{z}_k$ forward one 64-token window at a time with a GRU, starting from the last memory state and without generating tokens. D-HIER first forms a coarse query for the whole remainder, then forms fine window queries conditioned on the coarse vector.

Hazard head. A logistic hazard $h_k = \sigma(\boldsymbol{w}^\top \boldsymbol{f}_k)$ gives the probability that the event first occurs in window k, and the cumulative risk within horizon H is

$$F(H) = 1 - \prod_{k < H/64} (1 - h_k). \tag{1}$$

We train it with the censored discrete-time likelihood of Gensheimer and Narasimhan [2018]: an event in bin j contributes $\log h_j + \sum_{i<j} \log(1 - h_i)$, and a trajectory censored in bin j contributes $\sum_{i<j} \log(1 - h_i)$.

Content heads and JEPA auxiliary. Behavior, topic and summary heads read $\boldsymbol{f}_k$ (or $\boldsymbol{z}_k$). The JEPA-style auxiliary is a cosine loss between a linear map of $\boldsymbol{f}_k$ and a PCA-128 projection of the target model's own mean activation over future window k.

Matched direct readout. DIRECT-AO reads only the latest block and passes it through an MLP ($256 \to 1024 \to 1024 \to 256$). It predicts safety with six independent sigmoids, one per $H \in \{64, 128, 256, 512, 1024, 2048\}$, trained with masked binary cross-entropy, and uses the same content heads.

Ablations and controls. We test A-QUERY without memory (latest block only), without the JEPA loss, and with independent per-H classifiers on pooled forecasts instead of the hazard head, as well as B-ROLLOUT without memory. At test time we also replace activations with zeros or with activations shuffled across trajectories.

3.3 Baselines and metrics

Non-neural baselines. *Prevalence* is a censoring-aware Kaplan–Meier hazard fit on train. *Position* is logistic regression on the cutoff index. *Linear probe* is L2-regularized logistic regression on a PCA-128 projection of the latest block, one per H. *Text-only* is TF-IDF (1–2-grams) of prompt and prefix text with logistic regression per H. For content, *majority* predicts the most frequent label at each distance and *copy-last* repeats the annotator's label of the last observed segment, which makes

Table 1: Trajectories per prompt source. Event columns give the number of trajectories with the event (in parentheses: with onset $\geq$64 tokens, which can contribute positives at $H > 64$). Unsafe onsets occur at 0, 64, 128, 256 and 448 tokens only.

Source	n	Test	Median len.	Ended EOS	Unsafe	Caution
LONGWRITER	32	9	1559	21	2 (2)	3 (3)
TOXICCHAT	16	7	116	16	0	8 (0)
BEAVERTAILS	16	4	127	16	2 (1)	12 (0)
AEGIS-2.0	16	6	125	16	1 (1)	12 (2)
XSTEST	16	2	242	16	0	7 (2)

Table 2: Unsafe-onset forecasting: AUROC on known-status still-safe test prefixes (mean $\pm$ sd over 3 seeds). Test positives per H are 2/4/6/10/10/10 prefixes from only 2 event trajectories, so this event is unpowered; no observer differs significantly from DIRECT-AO. Best per column in **bold**.

Method	H=64	128	256	512	1024	2048
Prevalence (KM)	0.500	0.500	0.500	0.500	0.500	0.500
Position	0.697	0.729	0.704	0.751	0.685	0.900
Linear probe	0.658	0.710	0.771	0.841	0.820	**0.933**
Text-only TF-IDF	**0.732**	**0.816**	**0.855**	**0.929**	**0.920**	**0.933**
DIRECT-AO	0.530$\pm$.04	0.527$\pm$.01	0.710$\pm$.01	0.805$\pm$.02	0.825$\pm$.01	0.878$\pm$.02
A-QUERY	0.502$\pm$.04	0.494$\pm$.03	0.668$\pm$.02	0.758$\pm$.04	0.777$\pm$.04	**0.933**$\pm$.00
B-ROLLOUT	0.514$\pm$.03	0.599$\pm$.05	0.731$\pm$.03	0.795$\pm$.05	0.808$\pm$.04	0.889$\pm$.07
D-HIER	0.531$\pm$.13	0.466$\pm$.13	0.577$\pm$.17	0.618$\pm$.21	0.600$\pm$.22	0.811$\pm$.10

it a strong, partly oracle baseline. We make every per-H classifier monotone in H with a cumulative maximum.

Safety metrics. Per horizon H we report AUROC, AUPRC, Brier score and log loss on known-status prefixes: a prefix is excluded if it is censored before H. We compare each observer to DIRECT-AO using seed-averaged risk and 500 family-cluster bootstrap resamples. For early warning, an alarm fires when $F(512)$ exceeds the 95th percentile of the maximum risk over test trajectories without the event (a 5% trajectory false-alarm rate).

Content metrics. We report behavior accuracy (7 classes) and topic-cluster accuracy (16 classes) at distance k segments, and summary retrieval MRR and top-10 accuracy among all 330 test cutoffs.

4 Results

4.1 Natural onsets are rare and early

Table 1 summarizes the data. Only 5 of 84 continuations contain a segment flagged unsafe, and every unsafe onset is at or before token 448. Caution events are more frequent (42 trajectories), but most are refusals at token 0; only seven have a late onset (256, 384, 512, 768, 1152, 1152 and 1408 tokens). Segment behaviors are dominated by explain (614 segments), narrate (234) and recommend (133); refuse (27) and warn (23) are rare. Because 86% of responses end with EOS and are censored, the number of known-status still-safe test prefixes falls quickly with horizon: 306, 282, 239, 175, 89 and 13 for $H = 64, \ldots, 2048$ on the unsafe event. We therefore treat $H = 2048$ as uninformative throughout.

4.2 Unsafe-onset forecasting is unpowered

Table 2 reports AUROC for the primary event. The text-only TF-IDF baseline is best at every horizon (0.732–0.929 for $H \leq 1024$), and the linear probe and DIRECT-AO follow. None of the memory-based observers improves on DIRECT-AO, and every family-bootstrap 95% CI for the difference includes zero. Early-warning recall at a 5% trajectory false-alarm rate is 0 for every method, since only one test trajectory has an onset at or after 128 tokens. With 2 test event trajectories (onsets at 64 and 448 tokens), **we draw no architectural conclusion from this event**.

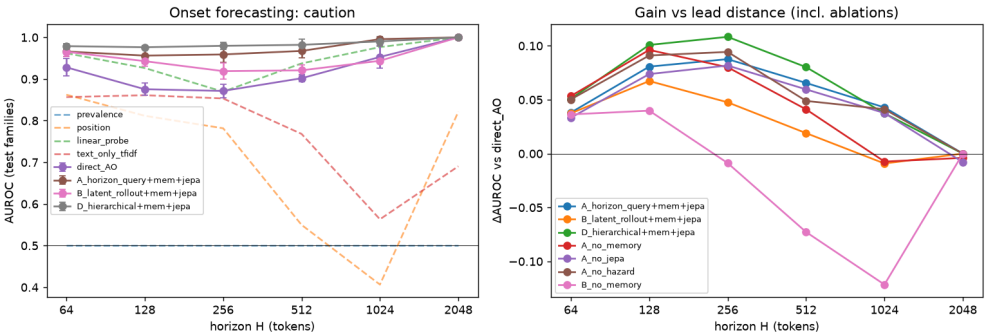


Figure 2: Caution-onset forecasting. *(Left)*AUROC vs. horizon for baselines (dashed) and observers (solid; error bars are seed sd). *(Right)*ΔAUROC vs. DIRECT-AO, including ablations. The memory-based observers lead at every $H \leq 1024$, but the gap peaks at $H = 128$–256 and shrinks by $H = 1024$; removing memory from B-ROLLOUT turns its gain negative at long horizons. All models reach 1.0 at $H = 2048$, where nearly all known-status prefixes are positive.

Table 3: Caution-onset forecasting (first refuse/warn segment). *Top:* AUROC on known-status still-safe test prefixes (mean ± sd over 3 seeds). *Bottom:* ΔAUROC vs. DIRECT-AO with seed-averaged risk and 95% family-cluster bootstrap CIs. H=2048 is omitted: only 31 known-status prefixes remain and nearly all are positive. Best per column in **bold**.

Method	H=64	128	256	512	1024
Position	0.862	0.812	0.782	0.550	0.406
Text-only TF-IDF	0.857	0.861	0.853	0.768	0.563
Linear probe	0.962	0.926	0.870	0.937	0.977
DIRECT-AO	0.928$_{\pm.02}$	0.875$_{\pm.01}$	0.871$_{\pm.02}$	0.902$_{\pm.01}$	0.953$_{\pm.03}$
A-QUERY (+memory, +JEPA, hazard)	0.966$_{\pm.01}$	0.956$_{\pm.02}$	0.959$_{\pm.02}$	0.967$_{\pm.02}$	**0.996**$_{\pm.00}$
B-ROLLOUT (+memory, +JEPA, hazard)	0.965$_{\pm.01}$	0.943$_{\pm.01}$	0.919$_{\pm.02}$	0.921$_{\pm.01}$	0.944$_{\pm.00}$
D-HIER (+memory, +JEPA, hazard)	0.979$_{\pm.01}$	**0.976**$_{\pm.00}$	**0.980**$_{\pm.01}$	**0.982**$_{\pm.01}$	0.990$_{\pm.01}$
A-QUERY − memory	**0.981**$_{\pm.01}$	0.972$_{\pm.01}$	0.951$_{\pm.01}$	0.943$_{\pm.01}$	0.946$_{\pm.01}$
A-QUERY − JEPA	0.961$_{\pm.02}$	0.949$_{\pm.02}$	0.953$_{\pm.02}$	0.961$_{\pm.01}$	0.991$_{\pm.01}$
A-QUERY − hazard head	0.978$_{\pm.00}$	0.967$_{\pm.01}$	0.966$_{\pm.00}$	0.951$_{\pm.02}$	0.994$_{\pm.01}$
B-ROLLOUT − memory	0.964$_{\pm.01}$	0.915$_{\pm.01}$	0.863$_{\pm.03}$	0.829$_{\pm.05}$	0.831$_{\pm.01}$
ΔAUROC vs. DIRECT-AO *[95% CI]*					
A-QUERY	+0.03 [−.05, +.07]	+0.08 [−.05, +.15]	+0.09 [−.03, +.16]	+0.07 [−.02, +.13]	+0.03 [.00, +.09]
B-ROLLOUT	+0.03 [−.01, +.05]	+0.06 [−.01, +.10]	+0.05 [−.04, +.10]	+0.01 [−.06, +.06]	−0.01 [−.07, +.01]
D-HIER	+0.04 [−.02, +.07]	+0.10 [−.01, +.15]	+0.11 [.00, +.16]	+0.09 [.00, +.14], p=.09	+0.03 [.00, +.09]
B-ROLLOUT − memory	+0.02	+0.03	−0.02	−0.08 [−.22, +.02]	−0.11 [−.20, .00], p=.06

4.3 Caution-onset forecasting

Do memory-based observers beat the direct readout? On the denser caution event, yes in sign, but not significantly (table 3, figure 2). D-HIER reaches AUROC 0.976–0.990 at $H \leq 1024$, compared with 0.871–0.953 for DIRECT-AO, a difference of $+0.04$ to $+0.11$. Its largest gains are at $H = 256$ ($+0.11$, CI [0.00, 0.16]) and $H = 512$ ($+0.09$, CI [0.00, 0.14], $p = 0.09$). A-QUERY gains $+0.03$ to $+0.09$. The gains are consistent across seeds, but every CI touches or includes zero, because only three test trajectories from two families have a late caution onset. A plain linear probe on the latest activation is nearly as strong (0.870–0.977), which shows that most of the signal is already present in the current state.

Does the gain grow with lead distance? No. The gap over DIRECT-AO peaks at $H = 256$–512 and shrinks to $+0.03$ at $H = 1024$ (figure 2, right). Part of this shrinkage is a ceiling: DIRECT-AO itself reaches 0.953 at $H = 1024$. D-HIER has the flattest curve of all models.

Which components matter? *Memory* gives the clearest long-horizon benefit. Removing it from B-ROLLOUT lowers AUROC from 0.921 to 0.829 at $H = 512$ and from 0.944 to 0.831 at $H = 1024$ (Δ vs. DIRECT-AO $= -0.11$, $p = 0.06$). Removing it from A-QUERY lowers AUROC at $H = 1024$ from 0.996 to 0.946. At short horizons memory does not help: A-QUERY without memory is the best model at $H = 64$ (0.981). *The JEPA auxiliary* changes AUROC by about 0.01 at every horizon. *The hazard head* gives no AUROC gain over independent per-H classifiers (A-QUERY − hazard head is within 0.016 of A-QUERY everywhere). *Latent rollout* does not reach further than direct query-based prediction: B-ROLLOUT is the weakest memory-based observer at $H \geq 256$.

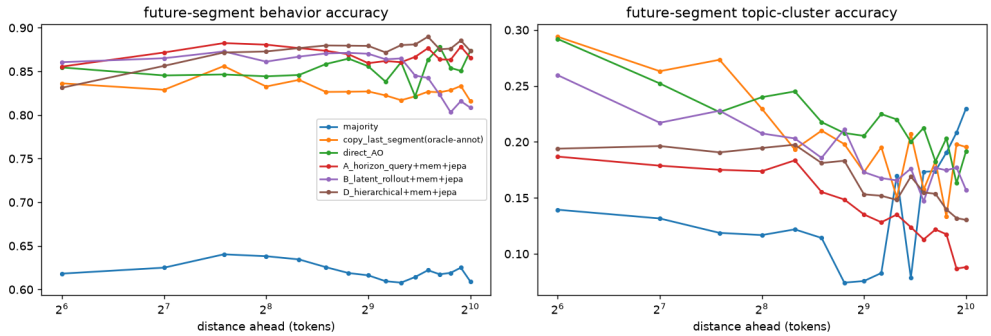


Figure 3: Content forecasting vs. distance ahead. *(Left)*Future-segment behavior accuracy: all observers stay at 0.83–0.89 up to 1024 tokens ahead, above the majority baseline (about 0.62) and slightly above copy-last-segment, which uses annotator labels. *(Right)*Topic-cluster accuracy: DIRECT-AO and copy-last are strongest; the query-based observers A-QUERY and D-HIER are weakest and decline with distance.

Calibration. At $H \leq 512$ all activation models have similar Brier scores (0.015–0.09, vs. 0.03–0.12 for prevalence). At $H = 1024$, DIRECT-AO scores 0.082 and A-QUERY without the hazard head 0.065, while the hazard models score 0.08–0.11 and B-ROLLOUT without memory 0.16. At $H = 2048$ the hazard models with memory degrade to 0.24–0.30, compared with 0.02–0.03 for independent per-H classifiers. They under-predict because equation 1 must extrapolate the hazard over late bins that are rarely observed in training.

Early warning. Only 3 test trajectories have a caution onset at or after 128 tokens. Recall at a 5% trajectory false-alarm rate is 0/3 for all observers except A-QUERY without JEPA (1/3), and 2/3 for TF-IDF; these counts are too small to compare methods.

4.4 Pooled AUROC largely reflects prompt type

Is the caution gain a timing forecast? Mostly not. At $H \geq 512$, all positive test prefixes for the caution event come from AEGIS-2.0 and XSTEST prompts, while LONGWRITER prefixes, the majority, are all negative. A model that separates harmful-looking prompts from long-form tasks therefore scores high without forecasting timing. Within a single source the evidence is thin (table 6 in appendix A). On the 9 XSTEST prefixes at $H = 512$, AUROC is 1.00 for D-HIER, 0.50 for B-ROLLOUT, 0.38 for DIRECT-AO and 0.00 for A-QUERY. On the unsafe event, within AEGIS-2.0 (14 prefixes, 8 positive), D-HIER and A-QUERY without JEPA reach 0.94 vs. 0.85 for DIRECT-AO, while within LONGWRITER (2 positives) every model is at or below chance (0.11–0.55). We therefore read pooled differences as "which observer better encodes prompt-type risk" rather than as evidence of long-range timing forecasts.

4.5 Long-form content forecasting

Can observers forecast what a long response will do? Yes, for behavior and summaries (table 4, figure 3). All activation observers forecast future-segment behavior at 0.83–0.88 accuracy up to 1024 tokens ahead, far above the majority baseline (0.61–0.64) and slightly above copy-last (0.82–0.84). Summary retrieval reaches MRR 0.074–0.101, compared with 0.019 for the no-activation baselines, a 4–5$\times$ improvement. A-QUERY and D-HIER beat DIRECT-AO on summary retrieval (MRR 0.095 and 0.099 vs. 0.087; top-10 0.235 and 0.239 vs. 0.200) and on behavior at 256–512 tokens ahead (0.873–0.881 vs. 0.844–0.856). DIRECT-AO is best on topic-cluster accuracy at $k \geq 4$, where A-QUERY falls to 0.088 at $k = 16$. Here memory is not needed: A-QUERY without memory has the best MRR (0.101).

Do observers actually read activations? Yes. Zeroing activations at test time drops behavior accuracy at $k = 4$ to the majority value (0.638) or below (0.06 for DIRECT-AO), and summary MRR to chance (0.019). Shuffling activations across trajectories gives behavior accuracy 0.46–0.52 and MRR 0.024–0.031. One caveat: for memory models with zeroed inputs, the padding mask

Table 4: Long-form content forecasting on all test cutoffs. Behavior accuracy (7-way) and topic-cluster accuracy (16-way) at distance k segments (64 tokens each), and remaining-response summary retrieval among 330 test cutoffs. Copy-last uses the annotator's label of the last observed segment. Seed sd is $\leq$0.02 for most cells (0.06 for B-ROLLOUT at k=16). Best per column in **bold**.

	Behavior acc.				Topic acc.				Summary	
Method	k=1	4	8	16	k=1	4	8	16	MRR	Top-10
Majority per distance	0.618	0.638	0.616	0.609	0.139	0.117	0.076	**0.230**	0.019	0.030
Copy last segment	0.836	0.833	0.827	0.816	**0.294**	0.230	0.173	0.195	0.019	0.030
DIRECT-AO	0.855	0.844	0.856	**0.874**	0.292	**0.240**	**0.205**	0.192	0.087	0.200
A-QUERY	0.856	**0.881**	0.859	0.866	0.187	0.174	0.135	0.088	0.095	0.235
B-ROLLOUT	0.861	0.861	0.870	0.808	0.260	0.208	0.173	0.157	0.074	0.160
D-HIER	0.831	0.873	**0.879**	**0.874**	0.194	0.195	0.153	0.130	0.099	0.239
A-QUERY − memory	0.829	0.869	0.872	0.862	0.238	0.202	0.202	0.176	**0.101**	0.249
A-QUERY − JEPA	**0.862**	0.873	0.859	0.870	0.195	0.180	0.135	0.100	0.094	0.239
A-QUERY − hazard head	0.860	0.879	0.870	0.870	0.178	0.157	0.139	0.092	0.098	**0.257**
B-ROLLOUT − memory	0.858	0.850	0.841	0.785	0.256	0.223	0.200	0.180	0.076	0.179

Table 5: Cost on one RTX A6000. About 4.2M parameters of each budget are the shared $16384 \to 256$ input projection.

Model	**Params**	**Train time / seed**	**Latency / 16-token update**
DIRECT-AO	6.00M	1 s	0.9 ms
A-QUERY	5.97M	3 s	2.3 ms
B-ROLLOUT	5.84M	4 s	3.8 ms
D-HIER	6.04M	3 s	2.5 ms

still reveals the cutoff position, which explains above-chance zero-control safety AUROC for some memory models.

4.6 Cost

All observers are cheap (table 5). Memory-based observers cost 2.3–3.8 ms per 16-token update, compared with 0.9 ms for DIRECT-AO, and train in 3–4 s per seed on our data.

5 Discussion

What the evidence supports. We tested four hypotheses. (H1) The memory-based observers beat DIRECT-AO only on the caution event, by $+0.03$ to $+0.11$ AUROC, with CIs that touch zero; the unsafe event cannot discriminate between methods. (H2) The gain does not grow with lead distance; it peaks at $H = 256$–512. (H3) Among components, memory helps at long horizons (-0.09 to -0.11 AUROC at $H \geq 512$ when removed from B-ROLLOUT), while the JEPA loss, the hazard head and latent rollout bring no measurable gain. (H4) Content forecasting beyond trivial baselines is supported: behavior is forecast at 0.87–0.88 accuracy 512–1024 tokens ahead, and summary retrieval reaches about 5$\times$ chance MRR.

Much of what is forecastable is prompt-level. Strong simple baselines are a recurring theme. A linear probe on the latest activation is competitive on the caution event, and text-only TF-IDF is the best unsafe-event predictor. Much of the behavior signal is response genre (e.g., narrate vs. explain), which persists through a response. These observations suggest that, at this scale, the forecastable content is mostly a prompt-level or genre property, readable from the prompt-end activation or prompt text, rather than a long-range dynamic plan. This is consistent with prior findings that prompt states encode global response properties [Dong et al., 2025] and that text-only baselines can rival activation readers [Bauer et al., 2026].

Why the hazard head did not help. In principle a hazard head gives monotone, coherent risk across horizons. In practice, with 86% of responses censored by EOS, late bins receive almost no supervision, and the product in equation 1 extrapolates poorly; its Brier score at $H = 2048$ is ten

times that of independent classifiers. A parametric tail, or a mixture of direct per-H heads with a monotonicity constraint, may keep the coherence without the extrapolation error.

Limitations. *Scale.* This is a pilot with 84 trajectories, 26 test trajectories in 12 test families, 5 unsafe and 7 late caution events; most CIs span ± 0.05–0.2. *Natural events are early.* No unsafe onset occurs after 448 tokens, so the 1024–2048 horizons are untestable for that event, and long-H labels come from a biased subset of long LONGWRITER outputs. *Caution is partly a refusal signal.* Most caution onsets are at token 0; those trajectories contribute only cutoff 0, and late onsets drive the long-H results. *Labels.* A single LLM annotator gives single-pass labels at 64-token resolution, with no inter-annotator check and no cross-validation against a guard model. *Baselines.* We did not run an off-the-shelf Activation Oracle or LatentQA decoder; DIRECT-AO is a matched MLP readout, and our "decoders" are classification and retrieval heads rather than free-text generators. *Other.* We used one rollout per prompt, so we cannot measure calibration against the model's true outcome distribution; hyperparameters were not tuned; and p-values are not corrected for multiple comparisons, which would only strengthen the null conclusions.

Safety considerations and broader impact. An early design that wrapped harmful requests in long-form templates to construct late unsafe onsets was abandoned, because it would have elicited detailed harmful content. We use only naturally occurring events from unmodified benchmark prompts, and we release no harmful generations. Better forecasting monitors could help stop harmful generations early; the same monitors could in principle be used to probe for prompts that delay detection, which is one reason we report their failure modes in detail.

6 Conclusion

We asked whether observers with trajectory memory, future-window queries, a future-activation loss and a censoring-aware hazard head forecast unsafe and long-form behavior of a frozen QWEN3-8B further ahead than a matched direct readout. In a pilot of 84 on-policy trajectories, the hierarchical and horizon-query observers rank caution onsets and retrieve remaining-response summaries somewhat better than the direct readout, but the differences are not significant, do not grow with lead distance, and largely reflect prompt-type separation. Memory is the one component with a consistent long-horizon benefit; the hazard head, JEPA loss and latent rollout did not earn their complexity. Long-horizon *unsafe* forecasting could not be tested, because unmodified prompts produce almost no late unsafe content.

The key takeaway is that a fair test of long-horizon activation forecasting needs data with late natural events and within-source evaluation, not just better architectures. Next steps are to scale data 10–$20\times$ with repeated rollouts per prompt, to seek natural late-onset regimes such as multi-turn agent transcripts or the transition from a reasoning trace to the final answer, to cross-check labels with a token-level guard model [Qwen Team, 2025], to add trained Activation Oracle and LatentQA decoder baselines, and to fix the hazard head's long-horizon calibration.

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A Per-source breakdown

Table 6: Per-source AUROC / Brier at $H = 512$ on known-status still-safe test prefixes (seed-averaged). "–" means AUROC is undefined because only one class is present. Pooled differences in table 3 are driven by between-source separation: all caution positives come from AEGIS-2.0 and XSTEST.

Source	n	pos	DIRECT-AO	A-QUERY	B-ROLLOUT	D-HIER	A-QUERY −mem	A-QUERY −JEPA	A-QUERY −hazard
Caution event									
AEGIS-2.0	16	16	–/0.485	–/0.444	–/0.509	–/0.464	–/0.478	–/0.417	–/0.540
BEAVERTAILS	2	2	–/0.000	–/0.000	–/0.000	–/0.000	–/0.000	–/0.000	–/0.000
LONGWRITER	145	0	–/0.003	–/0.000	–/0.004	–/0.000	–/0.000	–/0.001	–/0.003
TOXICCHAT	29	1	1.00/0.019	1.00/0.019	1.00/0.024	1.00/0.028	0.96/0.024	1.00/0.014	1.00/0.020
XSTEST	9	8	0.38/0.583	0.00/0.807	0.50/0.675	**1.00**/0.718	0.38/0.654	0.00/0.824	0.12/0.727
Unsafe event									
AEGIS-2.0	14	8	0.85/0.388	0.88/0.550	0.85/0.524	**0.94**/0.545	0.88/0.540	**0.94**/0.543	0.79/0.508
LONGWRITER	131	2	0.23/0.016	0.11/0.015	0.41/0.016	0.13/0.015	0.37/0.015	0.12/0.015	0.39/0.015
TOXICCHAT	28	0	–/0.000	–/0.000	–/0.000	–/0.000	–/0.000	–/0.000	–/0.000
XSTEST	2	0	–/0.035	–/0.000	–/0.000	–/0.000	–/0.001	–/0.000	–/0.006

B-ROLLOUT without memory scores 0.25 (caution, XSTEST) and 0.85 / 0.55 (unsafe, AEGIS-2.0/ LONGWRITER) on AUROC.

B Implementation and reproducibility

Hardware and time. All experiments run on one RTX A6000 (48 GB) with Python 3.12, PyTorch 2.14 and transformers 5.20. Generation took 423 s plus 108 s, activation caching about one minute, and observer training 2–4 s per model per seed. Annotation used 96 LLM calls costing under $1. An earlier generation run with 216 prompts ran out of GPU memory and was discarded; the final 96-prompt family split was again fixed before generation.

Annotation rubric. For each 64-token segment (QWEN3-8B tokenizer), the annotator returns a behavior from {explain, instruct, recommend, narrate, refuse, warn, other}, a 1–3 word topic, an unsafe flag and a description of at most 15 words, plus an overall response summary.

Leakage audit. Observer inputs are blocks that end at or before the cutoff, asserted in code; future-window activations are used only as JEPA targets; topic clusters and standardization statistics are fit on train only; and the family split is asserted disjoint.

Reproducibility check. Re-running DIRECT-AO and A-QUERY on the unsafe event with seed 0 reproduces DIRECT-AO exactly (AUROC 0.480 / 0.776 / 0.811 at H = 64 / 512 / 1024) and A-QUERY within seed noise (0.500 / 0.807 / 0.833 vs. the original 0.465 / 0.789 / 0.829); the differences come from non-deterministic CUDA attention kernels.